IVB Model 2 Documentation

Initial Balance breakout with order-flow confirmation — design, code map & contracts (updated 2026-06)

0. Overview

The IVB (Initial Balance Volume) model is a directional intraday futures strategy built on the Opening Range Breakout concept, enriched with volume-profile analysis and order-flow confirmation. Originally inspired by Fabio Valentini's IVB model.

Two layers: a **directional framework** (define the initial balance, detect a breakout, wait for a retest of the value area) shared across versions, and an **entry confirmation** layer where versions differ. The current production strategy is the package `strategies/ivb_model_2_folder_2/`, loaded by the backtester via its `__init__.run()`. It supports 30+ instruments via ASSET_INFO tick injection, direction flipping, a peak-based volume profile, a notes column, and — new — **four interchangeable absorption entry types**.

Layer	What it provides	How it is implemented
1. Direction	who won the most important battle of the day	first post-IB bar closing above ivb_high or below ivb_low
2. Location	where the reload zone is (value area)	VAH / POC / VAL from the IB tick_volume profile
3. Confirmation	the counter-move is failing at the reload zone	one of four absorption entry patterns (Section 4)

1. Package Structure & Cross-Connection

The strategy is a multi-file package (not a single file). Files and their contracts:

File	Provides	Needs / Returns
__init__.py	run(folder_path, start, end, params)	globs YYYY-MM-DD.parquet, calls process_day per day, returns OUTPUT_COLUMNS frame
params.py	PARAMS, PARAM_SECTIONS, OUTPUT_COLUMNS	default params, UI grouping, output schema
core.py	detect_breakout, detect_retest, find_entry, process_day	orchestrates one day; dispatches entry finders
profile.py	compute_ivb_profile(ib_bars)	reads tick_volume -> (poc, vah, val)
baselines.py	rolling / passive / two-bar baselines + merge_tick_volume	per-tick volume baselines for absorption grading
absorption.py	is_absorption_candle, find_absorption_trigger	shared wick-absorption grading from tick_volume
entries/	4 finders + FINDER_REGISTRY	each find_entry(...) -> 5-tuple
risk/	compute_sl_tp, run_trade (+ trailing.py)	SL/TP placement + fill simulation

Per-day call chain (process_day)

```
run() -> for each day file -> process_day(session, params):
  slice RTH 09:30-16:00 NY        (need >= ib_minutes bars)
  IB high/low from first ib_minutes bars        (abort if range <= 0)
  compute_ivb_profile(ib_bars)     -> poc, vah, val      [profile.py]
  build rolling + passive baselines (long & short)     [baselines.py]
  detect_breakout(post_ib)        -> direction, breakout_pos
  loop (<= max_flips):
     detect_retest()              -> retest_pos  (else no trade)
     post_retest = [breakout bar] + [retest .. retest+entry_window]
     find_entry()  -> earliest entry across enabled finders wins
        entry      -> break
        invalidate -> flip direction, resume after invalidation, re-detect breakout
  compute_sl_tp(post_retest)      -> sl, tp              [risk/sl_tp.py]
  run_trade(post_entry)           -> trade dict          [risk/sl_tp.py]
  attach trade_type + notes(JSON)
```

2. Directional Framework

Step 1 — Initial Balance

Scan the first `ib_minutes` bars of RTH (default 30 = 09:30–09:59). ivb_high = max high, ivb_low = min low. If the range is zero, skip the day. The 10:00 bar is the first post-IB bar.

Step 2 — Volume Profile

Compute POC/VAH/VAL from the IB bars' tick_volume using the peak-based algorithm (Section 5). Computed once per day and reused across all flips.

Step 3 — Breakout

First post-IB bar that **closes** above ivb_high (long) or below ivb_low (short); whichever comes first sets the direction. A wick that closes back inside is not a breakout.

Step 4 — Retest

Within `retest_window` bars after the breakout, wait for price to pull back into the value area — long: bar low ≤ VAH; short: bar high ≥ VAL. The entry scan window is the breakout bar plus the next `entry_window` bars from the retest.

Step 5 — Direction Flipping

Whole-trade invalidation = a close back through VAL (long) / VAH (short). On invalidation the strategy flips direction, resumes scanning after the invalidation bar, and re-detects a breakout in the new direction — up to `max_flips` times. The same IB profile is reused. If the re-detected breakout is the wrong direction, no trade.

3. Absorption Grading & Baselines

is_absorption_candle (absorption.py): true when the defended wick (lower for long, upper for short) is ≥ `wick_threshold` of the bar range AND some price level inside that wick has counter-trend aggressive volume ≥ baseline × `absorption_mult`. **find_absorption_trigger** returns the first qualifying (price, volume) for the notes.

- **build_rolling_baseline** — sell/buy volume-per-tick, rolling mean over absorption_baseline_window bars, from 09:35 (first 5 min skipped). Used by absorption_delta, consecutive, passive.
- **build_passive_baseline** — rolling mean of max(size/count) on the defended side per bar (below open for long, above for short). Used by passive_absorption.
- **build_two_bar_baseline** — average per-tick defended-side volume of merged 2-bar pairs preceding the pattern. Used by two_bar_absorption.

4. Entry Types (entries/)

All four share the signature and return `(entry_ts, entry_price, invalidation_ts, entry_notes, trade_type)`. The dispatcher runs every finder enabled by the `valid_entries` flag string ("1111", one bit per finder in FINDER_REGISTRY order) over the same window and baselines, then takes the **earliest entry**; if none, the earliest invalidation (which drives a flip). Entry is always the **open of the bar after** the confirming candle.

trade_type	Pattern	Key params
absorption_delta	absorption candle, then a confirming candle (correct direction, body ≥ body_threshold, |volume_delta_pct| ≥ delta_threshold)	wick_threshold, absorption_mult
consecutive_absorption	consec_abs_n absorption candles clustered within consec_abs_ticks of the same level/body-mid; no delta confirmation	consec_abs_mult, consec_wick_threshold
two_bar_absorption	reversal pair (small wicks ≤ two_bar_wick_ticks) merged into a synthetic candle, graded vs the 2-bar baseline, then a confirming candle	two_bar_abs_mult, two_bar_wick_ticks
passive_absorption	big resting order on defended side (size/count ≥ passive_baseline × passive_order_mult) AND absorption on the same candle, then a confirming candle	passive_order_mult, passive_absorption_mult, passive_wick_threshold

passive_orders is now required. The passive_absorption finder consumes the passive_orders column directly, and build_passive_baseline reads it too. Earlier documentation listed passive_orders as unused — that is no longer the case.

Adding a fifth entry type

- Drop entries/my_entry.py exposing find_entry(**shared) -> 5-tuple.
- Append it to FINDER_REGISTRY in entries/__init__.py.
- Extend the default valid_entries string by one bit and add any params to params.py.

5. Volume Profile Algorithm (peak-based)

- Aggregate tick_volume across IB bars into {price: total_volume}.
- Smooth (3-tick rolling avg); find local maxima; cluster within 4 ticks; top 5 clusters.
- Refine each cluster to the highest RAW tick within ±3 -> POC candidate.
- Expand each candidate to a 70% value area; pick the tightest VAH-VAL range.
- Refine the POC to the highest raw-volume tick inside the winning value area.
- Falls back to a simple max-volume POC when fewer than 3 price levels exist.

6. Risk Management (risk/)

Stop loss / take profit — compute_sl_tp

`sl_type=0`: SL at VAL (long) / VAH (short). `sl_type=1`: swing low/high of bars up to entry. risk = |entry-sl|; tp = entry ± risk × `rr`. Returns (None, None) if risk is non-positive (day skipped).

Trade simulation — run_trade

Phase 1 (first `trade_timeout` bars): if TP and SL are both touched in the same bar, TP wins. Phase 2 (timeout reached): if in profit, exit at the timeout close (tp_timeout); if in loss, move TP to breakeven and SL to the swing extreme, then run on. Phase 3: exit on the adjusted SL/TP or at end of session (eod).

exit_reason	Meaning
tp / sl	take-profit / stop touched within the timeout window
eod	session ended without hitting a level
tp_timeout	in profit at timeout, or adjusted TP (breakeven) hit afterwards
sl_timeout	adjusted SL (swing) hit after the timeout

`risk_script` is a parameter reserved for selecting a risk module; `risk/__init__.py` currently always uses sl_tp, and `trailing.py` is a placeholder for a future trailing stop.

7. Parameters

Defaults from `params.py`; `PARAM_SECTIONS` groups them in the backtester UI.

Parameter	Default	Description
ib_minutes	30	IB duration in bars (15 / 30 / 60)
trade_timeout	999	bars before timeout logic (999 = effectively disabled)
max_flips	4	max direction flips per day after invalidation
valid_entries	"1111"	enable/disable each finder (order = FINDER_REGISTRY)
risk_script	1	reserved risk-module selector (currently always sl_tp)
retest_window	30	max bars to wait for a retest after breakout
entry_window	15	bars scanned for entry after the retest
entry_after_absorption	5	bars scanned for a confirming candle after absorption
absorption_baseline_window	20	rolling N bars for baselines (shared)
delta_threshold	30.0	min |volume_delta_pct| for a confirming candle

Parameter	Default	Description
body_threshold	0.5	min body/range ratio for a confirming candle
wick_threshold	0.4	min wick/range ratio (absorption_delta)
absorption_mult	2.0	wick-level volume must be >= this x baseline
consec_abs_n	2	absorption candles required at the same level
consec_abs_mult	2.0	absorption multiplier (consecutive)
consec_abs_ticks	4	level tolerance in ticks (consecutive)
consec_wick_threshold	0.4	wick threshold (consecutive)
two_bar_wick_ticks	3	max defended wick in ticks on both bars
two_bar_abs_mult	2.0	absorption multiplier for the merged pair
passive_order_mult	3.0	passive size/count must be >= this x passive baseline
passive_absorption_mult	1.5	absorption multiplier (passive)
passive_wick_threshold	0.4	wick threshold (passive)
rr	1.0	fixed risk:reward ratio
sl_type	0	0 = VAL/VAH, 1 = swing low/high
tick_size	(auto)	injected from ASSET_INFO; in HIDDEN_PARAMS, never set manually

8. Output Schema

The strategy returns one row per trade with the columns below (`OUTPUT_COLUMNS`). The backtester derives ticks from pnl_points — the strategy never stores ticks.

Column	Type	Description
date	date	trade date
direction	str	lowercase 'long' / 'short'
trade_type	str	which finder fired (absorption_delta / consecutive_absorption / two_bar_absorption / passive_absorption)
entry_time / exit_time	Timestamp	entry bar open time / exit bar time
entry_price / exit_price	float	open of entry bar / SL,TP, or close at exit
sl / tp	float	stop-loss / take-profit levels used
exit_reason	str	tp / sl / eod / tp_timeout / sl_timeout
pnl_points	float	exit-entry (long), entry-exit (short)
notes	JSON str	see Section 9

9. Notes Column

process_day context merged with finder-specific keys, serialized as JSON and rendered by the backtester as a second metrics row.

```
{
  "breakout_time": "10:03",   "retest_time": "10:11",
  "flip_count": 0,
  "ivb_high": 5102.25, "ivb_low": 5094.50,
  "poc": 5097.75, "vah": 5100.25, "val": 5095.50,
  // + finder-specific, e.g.:
  "absorption_time": "10:14", "trigger_price": 5096.0, "trigger_volume": 412
  // passive: passive_price/size/count ; two_bar: two_bar_baseline ; etc.
}
```

10. Data Requirements

Requires enriched 1-minute candles from `transforms/1m_advanced.py` (ES/NQ only). Plain OHLCV is not sufficient. Index must be tz-aware (America/New_York).

Column	Required	Used for
open / high / low / close	Yes (float64)	IB range, breakout/retest, SL/TP, bar structure
buy_volume / sell_volume	Yes	rolling absorption baseline (per-tick)
volume_delta_pct	Yes	confirming-candle delta threshold
tick_volume	Yes	volume profile + absorption grading
passive_orders	Yes	passive_absorption finder + passive baseline
volume	No	present but not used directly by the core logic

11. Performance

Prior performance / Monte-Carlo tables described the legacy single-absorption-entry model and no longer reflect the current four-entry version (which adds the trade_type dimension). They have been removed pending a fresh backtest. Re-run the backtester on ES enriched 1-minute candles and break results down by trade_type before quoting numbers here.

12. Known Issues & Roadmap

- **Session times** hardcoded 09:30–16:00 NY — wrong for non-equity assets.
- **Data coverage**: enriched tick_volume/passive_orders only for ES/NQ over one year — thin sample; walk-forward needs multi-year data.
- **Baseline warmup**: absorption baselines need a full window (from 09:35), so early candles cannot be graded.
- **Flip conservatism**: a re-detected breakout in the wrong direction yields no trade.
- **No higher-timeframe context**: operates purely on 1-minute bars.

Item	Status
Peak-based volume profile + POC refinement within VA	DONE
Absorption + delta entry	DONE
Consecutive absorption entry	DONE
Two-bar (merged reversal) absorption entry	DONE
Passive-order + absorption entry	DONE
trade_type column + per-finder notes	DONE
Direction flipping, notes column, HIDDEN_PARAMS tick injection	DONE
Per-asset session times	TODO
Trail SL to absorption level (risk/trailing.py)	TODO
Minimum-risk threshold -> extend TP when SL too close	TODO
Walk-forward validation (multi-year enriched data)	TODO